

Investment Strategy Challenge 2026 · Proposal for NKE Private Wealth (Chak family)

Asian Digital Transformation Autocallable Note

12.75% a year from Asia's semiconductor build-out, paid each quarter the basket is at or above its start

USD 100mn · 5-year note issued by Natixis · 12.75% p.a. memory coupon · autocall from Year 1 ·
Trade date 17 Sep 2026

Agenda

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Client & market

Who the Chak family is, what they need, and why Asia's digital build-out now

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Investment thesis & strategy

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Underlying basket, term sheet, and why a classic memory-coupon autocallable

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Risks & hedging

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Executive summary

One note that turns the family's view on Asian AI hardware into double-digit income

12.75%

Memory coupon p.a.

3.19% each quarter the basket is $\geq$ 100%; missed coupons are paid later when it recovers

0.8%

Chance of a loss

On the team outlook (slides 4–5). If markets go nowhere it rises to 33%: never called, repaid at the basket level

78%

Called at first review

Team outlook. 99% called within 5 years; expected life 1.2 years

We recommend **a USD 100mn, 5-year Natixis autocallable note** on a basket of five Bloomberg-listed Asian ETFs: Global X Asia Semiconductor (3119 HK) 35%, iShares MSCI South Korea (EWY) 20%, iShares MSCI Taiwan (EWT) 15%, Global X Japan Semiconductor (2644 JP) 15% and Cathay Taiwan ESG High Dividend (00878 TT) 15%. It gives the family the AI hardware layer (chips, memory and chip-making equipment) that complements, rather than duplicates, the infrastructure, power and property they already own. **On the team's outlook (slides 4–5) the note is called in 99% of paths, usually within 1.2 years, paying 12.75% a year; if the basket never gets back to its start (0.8%; 33% if markets go nowhere), the note repays the basket's level at Year 5. Fair value is 96.5%, so issuing at par leaves Natixis 3.5% for hedging and margin.**

The family wants growth from Asia's AI build-out, on 5% of its wealth

Client diagnosis: third-generation dynasty, succession done, now allocating for legacy

Who they are

- Single-family office: NKE Private Wealth
- Net worth > USD 2bn; mandate USD 100mn (~5%)
- Businesses: tech infrastructure, renewable power plants, real estate in Japan and Greater China

What they want

- Capital appreciation for future generations
- Meaningful exposure to Asia's digital transformation
- To bridge traditional Asian sectors with emerging tech they know

What they can bear

- Moderate risk: income first, capital usually back early
- 5-year horizon; coupons can be reinvested
- Accepts limited liquidity and equity risk on 5% of wealth

Client need	Design answer in our note
Capital appreciation	12.75% p.a. coupon, above the 5.69% 5Y USD funding rate, in every path that autocalls; proceeds roll into the next note
Exposure to Asian digital transformation	Semiconductor-heavy basket: the AI hardware layer behind the family's tech-infrastructure business (slide 6)
Generational preservation	Diversified basket at the low-risk edge of all mixes (slide 4); capital back early in 99% of paths; BPCE-backed issuer
Volatile markets	Memory coupon: a missed coupon is paid when the basket recovers, so a temporary drawdown costs nothing

Why now: Asia builds the hardware of the AI economy

Market view: structural growth themes, and high rates and volatility that pay for a double-digit coupon

1 Taiwan: the AI foundry

TSMC and its supply chain make most of the world's advanced AI chips; Taiwan's index is dominated by semiconductors.

2 Korea: memory for AI

Samsung and SK hynix supply the high-bandwidth memory every AI accelerator needs; together ~44% of EWY.

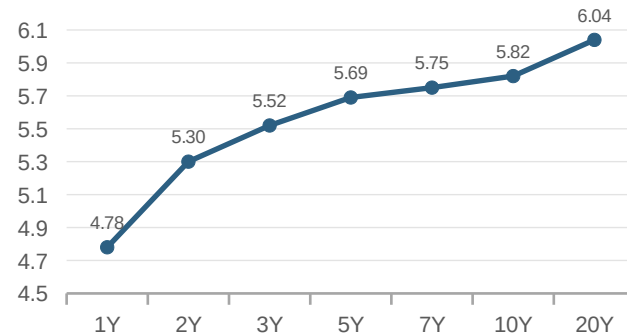
3 Japan: chip-making tools

Tokyo Electron, Advantest and peers make the equipment and testers chip factories need, alongside TSE governance reforms.

4 ESG sleeve

00878 adds ESG-screened Taiwan leaders (MSCI ESG methodology) and a lower-volatility, high-dividend anchor to the basket.

USD funding curve (case grid, %)



What the curve means for structuring

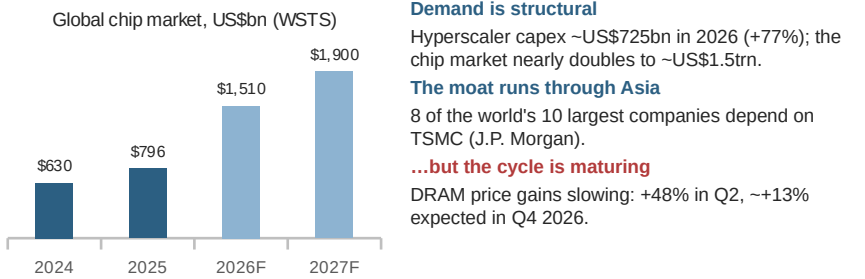
High USD rates (5Y funding 5.69%) and Asian volatility (basket ~26%) make option premium rich, and the basket is likely to revisit 100%: together they pay the 12.75% coupon.

Thesis is qualitative; index-level data to be refreshed on Bloomberg. Funding grid: Investment Strategy Challenge 2026 rules.

Investment thesis: Monopolizing the AI Supply Chain via Structured Overlays

What we own and why we wrap it: structural demand for Asian chips, memory and tools, held through a market of high rates, record volatility and concentrated, full prices

1 Asia controls AI's hardware bottlenecks



~70%

Taiwan makes the chips: TSMC share of global foundry revenue; TSMC >40% of TAIEX

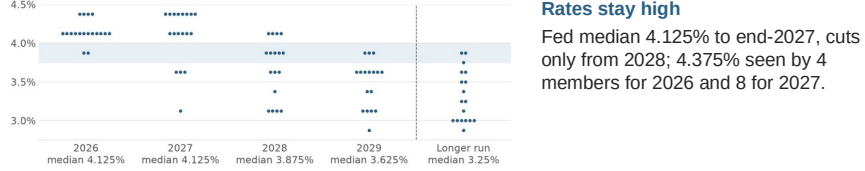
83%

Korea makes the memory: SK hynix + Samsung share of HBM, the memory every AI accelerator needs

~55%

Japan makes the tools: Advantest share of chip test equipment; with Tokyo Electron ≈ 20% of the Nikkei

2 2026 punishes bonds and buy-and-hold



FOMC dot plot, 16 Sep 2026 (shaded: today's 3.75–4.00%)

–4.0%

Long bonds carry rate risk: a 5-year bond loses ~4.0% per 1% rise.
→ Redeems from year 1 (expected life ~2 years): short duration, cash back to reinvest

91.2

Volatility is at records: VKOSPI's all-time high; KOSPI fell ~25% from June to August 2026.
→ Coupons pay on recovery, no timing needed; rich option premium funds them

~92%

Concentrated and fully priced: of advanced chips come from Taiwan; Nikkei CAPE 38.6×.
→ Diversified five-ETF basket; coupons lock in gains if the rally stalls

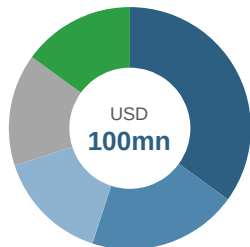
Whichever AI company wins, it needs Asia's chips, memory and tools; a memory-coupon autocall pays while the market swings.

Sources: WSTS; company capex guidance; Counterpoint (foundry, HBM); TrendForce (DRAM); industry research (test equipment); J.P. Morgan AWM 2026 Outlook; Federal Reserve SEP (16 Sep 2026, dots read from chart); Herald / SBS (VKOSPI); FN News, BIT Research (KOSPI); Sibilis (CAPE); case USD funding grid. Full list: deck/thesis_sources.md.

Strategy: our five-ETF basket sits at the low-risk edge of every possible mix

USD 100mn across five Bloomberg-listed ETFs; our weights (red star) deliver ~26% expected return for only ~17% volatility

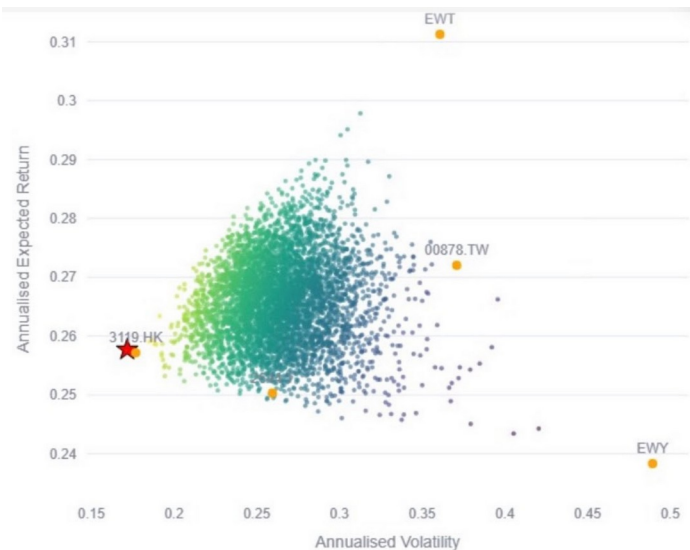
Allocation



Half in the semiconductor core
3119 HK + 2644 JP = 50%; Korea adds memory, Taiwan the foundry, 00878 the ESG income.

■	3119 HK	Pan-Asian chip leaders	35%
■	EWY US	Korea: memory for AI	20%
■	EWT US	Taiwan: the AI foundry	15%
■	2644 JP	Japan: chip-making tools	15%
■	00878 TT	ESG dividend anchor	15%

Risk vs return: single ETFs and random mixes



How to read it

Orange dots: each ETF on its own (2644 is the dot near 0.26, 0.25).

Red star: our basket (35 / 20 / 15 / 15 / 15).

Small dots: thousands of random weightings of the five ETFs; lighter = more return per unit of risk.

Axes: annualised volatility (risk) and expected return.

Takeaways

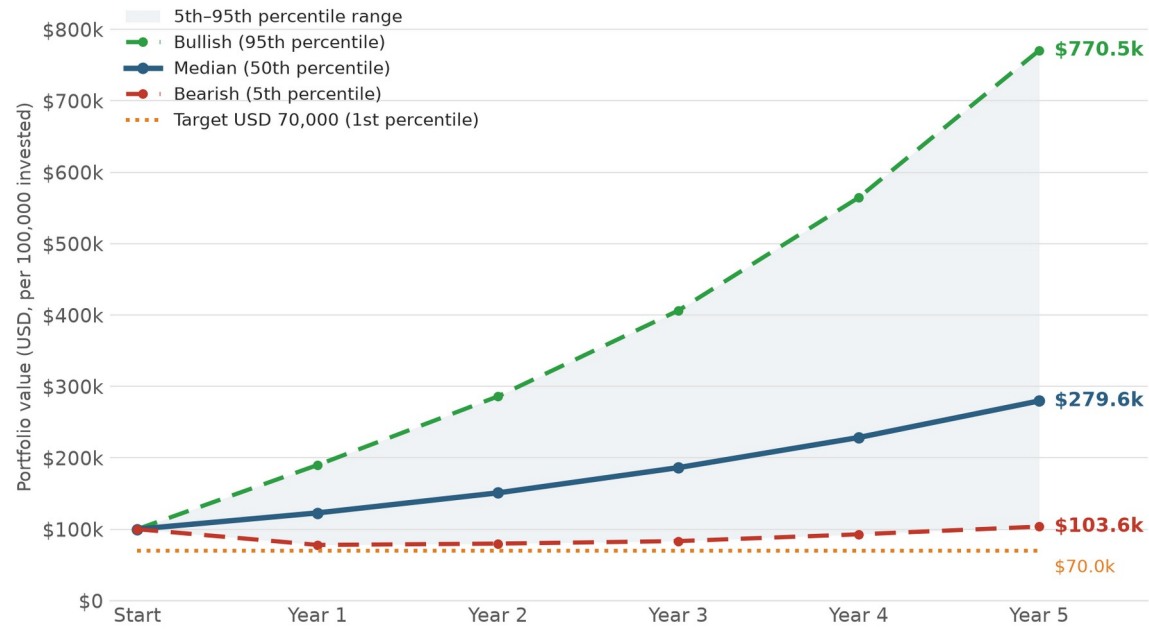
- Our basket (red star): ~26% return at ~17% volatility, the lowest risk of all the mixes shown.
- Most random mixes need 22–32% volatility for the same 25–28% return: our weights get it with less risk.
- Less risk than any single ETF, about a third of EWY's (~49%), while giving up only a few points versus EWT's ~31%.
- Why: 3119 HK (35%) anchors the basket; the other four move differently enough to pull risk down.

Our basket (red star): ~26% return for ~17% volatility, less risk than any single ETF or random mix

Risk/return chart: team analysis app (annualised from historical data; red star = our basket at 35/20/15/15/15; values read from the chart). Liquidity: 3119 HK is borderline against the USD 5M/day rule, verify on Bloomberg.

Monte Carlo: the median portfolio grows 2.8× in five years

5-year simulation of the five-ETF portfolio, per USD 100,000 invested (×1,000 for the USD 100mn mandate)



\$279.6k

Median at Year 5: 2.8× the start, +22.8% a year

\$103.6k

Bearish (5th percentile): still above the start, +0.7% a year

\$770.5k

Bullish (95th percentile): 7.7× the start

99.0%

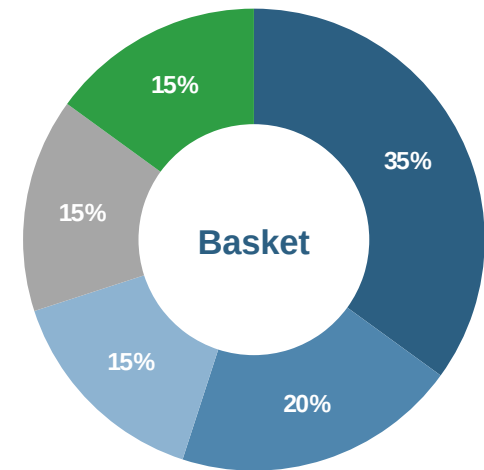
Chance of ending at or above the USD 70,000 target

Even the bearish path ends above the start (\$103.6k); the USD 70,000 target is the 1st percentile

Source: team Monte Carlo app (annual checkpoints of the simulated paths). Lines connect the yearly values. Results depend on the app's return and volatility inputs; refresh with Bloomberg data at the 17 Sep 2026 trade date.

Underlying: five Asian ETFs across the AI-hardware chain

Bespoke weighted basket, USD payout (quanto on the HKD, JPY and TWD lines); all Bloomberg-listed



- Asia Semiconductor (3119 HK)
- iShares MSCI South Korea (EWY)
- iShares MSCI Taiwan (EWT)
- Japan Semiconductor (2644 JP)
- Taiwan ESG High Dividend (00878 TT)

ETF (Bloomberg)	Wt	Role in the basket	Vol*
Asia Semiconductor (3119 HK)	35%	Pan-Asian chip leaders (40 names; TSMC, Samsung, SK hynix the largest): the core AI-hardware exposure	18%
iShares MSCI South Korea (EWY)	20%	Korea large caps (Samsung + SK hynix ~44%): high-bandwidth memory for AI	49%
iShares MSCI Taiwan (EWT)	15%	Taiwan large caps (TSMC, MediaTek, Delta): AI foundry and server chain in the family's Greater China region	36%
Japan Semiconductor (2644 JP)	15%	Japanese semiconductor stocks (FactSet Japan Semiconductor Index, e.g. Tokyo Electron, Advantest): the family's home market	26%
Taiwan ESG High Dividend (00878 TT)	15%	ESG-screened Taiwan high-dividend leaders (Quanta, UMC, MediaTek): ESG sleeve and lower-volatility anchor	37%

Why a weighted basket, not worst-of: a shock to one ETF is cushioned by the others: basket vol ~26% vs 18%–49% alone. Overlap (TSMC, Samsung, SK hynix sit in several ETFs) keeps correlations high (0.45–0.85).

*Annualised vols from the team risk/return app (slide 4). Basket vol ~26% with Bloomberg-style correlations, the level the slide-5 Monte Carlo implies (the app's chart estimates ~17%). USD 5M/day ETF rule: EWY, EWT, 2644, 00878 pass; 3119 is borderline (~USD 4–5M/day), verify the 6-month average on Bloomberg. 00878 distributes ~7.6% a year, which lowers its price path.

Term sheet: 5-year autocallable, 12.75% memory coupon

Why this design: the highest coupon on the basket, with capital usually back early

Issuer	Natixis SA (BPCE Group)
Notional / currency	USD 100,000,000 · quanto USD
Trade / maturity	17 Sep 2026 / 17 Sep 2031 (5Y)
Underlying	Weighted basket of 5 Asian ETFs (slide 6)
Observation	Quarterly, 20 dates; autocall from Q4 (Year 1)
Coupon	12.75% p.a. (3.19% per quarter), paid if basket $\geq$ 100%
Memory	Missed coupons paid on the next date basket $\geq$ 100%
Autocall	Basket $\geq$ 100% from Q4: 100% + coupons due, note ends
At maturity	Not called: repaid at the basket level (no floor)
Issue price / fair value	100% / 96.5%

Autocall design (same basket, issued at 98%)	Fair cpn	Loss risk (pricing)	Max loss
Classic fixed coupon, 70% barrier	7.2%	15.8%	Up to 100%
Snowball, 65% barrier	13.6%	14.0%	Up to 100%
Memory coupon, 70% barrier + 70% put	11.2%	15.8%	30%
Memory coupon, 100% protected	7.9%	0.0%	None
Classic memory autocall, no barrier (ours)	14.1%	21.6%	Up to 100%

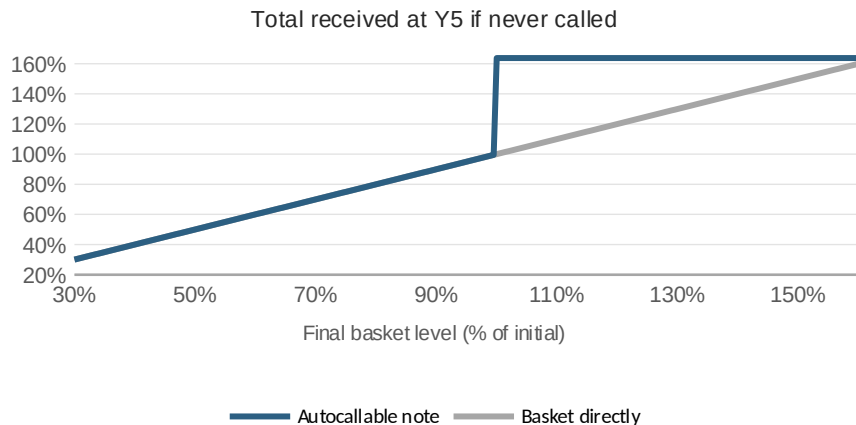
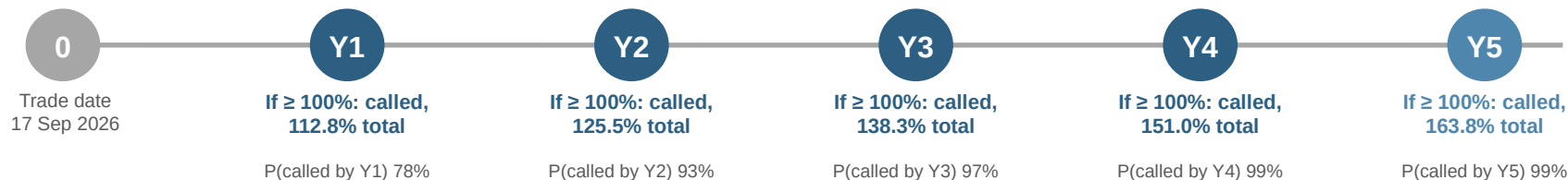
Design call: The draft's 8.5% fixed coupon is worth 100.6%, so Natixis cannot fund it. Full protection pays only 7.9%; a 70% floor pays 11.2%. Without barrier or floor the classic autocall is worth 14.1%: the family takes the basket's fall if the note is never called. We offer 12.75%.

Monte Carlo, 200k paths. Vols, dividends and correlations are assumptions to refresh with Bloomberg data as of the 17 Sep 2026 trade date; discounting uses the USD funding grid. Loss risk = probability the note returns less than 100% at maturity (risk-neutral).

How it pays: one question every quarter, three outcomes

Is the basket at or above 100% of its starting level? (checked quarterly; one date per year shown)

Q1–Q3: coupon only (3.19% if basket $\geq 100\%$). From Q4: coupon plus autocall. Missed coupons are always caught up.

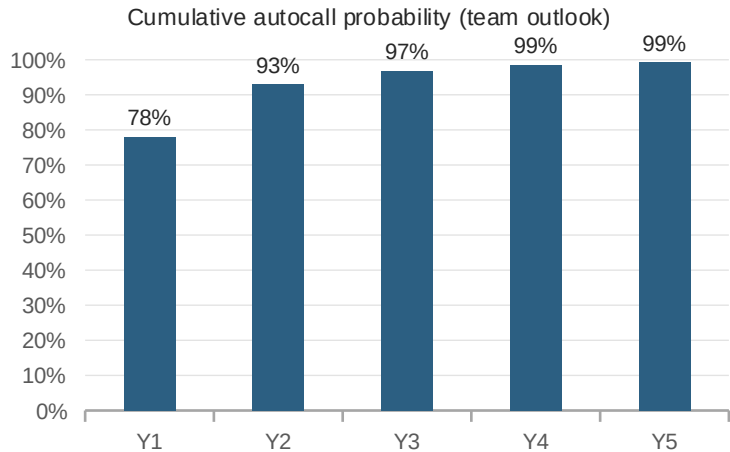


A	Autocalled Basket $\geq 100\%$ on any date from Q4: 100% plus every coupon to date. The note ends.	99.3% team outlook
B	Not called, basket 70–100% at Y5 Repaid at the basket level (e.g. 85% back), plus any early coupons.	0.6% team outlook
C	Not called, basket < 70% at Y5 Repaid at the basket level: the family bears the full fall.	0.2% team outlook

Probabilities on the team outlook (app returns and vols, slides 4–5); other scenarios and the pricing measure are on slide 9. The desk's replication (bond, coupon digitals, short put) is in Appendix C; stress paths in Appendix B.

Monte Carlo: on the team outlook the note is called in 99% of paths

Note outcomes on the team outlook (slides 4–5), on half that outlook, if markets go nowhere, and under the pricing measure



Metric	Team outlook	Half outlook	0% (flat)	Pricing (RN)
Called within 5Y	99%	91%	67%	78%
Expected life (yrs)	1.2	1.7	2.6	2.3
Mean IRR	13.0%	11.1%	4.3%	7.8%
Median IRR	13.3%	13.2%	12.6%	12.9%
Avg coupons received	15.2%	16.6%	13.4%	15.4%
Avg repayment if not called	79%	67%	51%	58%
Capital loss (not called)	1%	9%	33%	22%

78%

called at the first review
(112.8% total)

13.3%

median return a year on
the team outlook

33%

chance of a loss if
markets go nowhere

Monte Carlo, 200k paths. Vols, dividends and correlations are assumptions to refresh with Bloomberg data as of the 17 Sep 2026 trade date; discounting uses the USD funding grid. IRR uses actual quarterly coupon dates.

Risks for the family, and how Natixis hedges its side

Transparent disclosure plus a desk hedging plan for every exposure

Risk	What could happen	Mitigant
Capital loss	Never called: repaid at the basket level (0.8% on the team outlook; 33% if markets go nowhere)	Diversified basket, memory coupon; only 5% of wealth
Capped upside	Team median is 2.8× in 5 years; the note pays 12.75% p.a. and is usually called by Y1	Coupons lock in gains if the rally stalls; rest of family wealth keeps direct equity exposure
ETF structure	00878 pays ~7.6% a year out, lowering its price path; 3119 trading is thin; ETF tracking error	00878 held at 15%; verify volumes; Natixis can reference total-return versions
Issuer credit	Coupons and repayment depend on Natixis / BPCE	BPCE senior rating; optional collateralised wrapper
Liquidity / MTM	Sold before maturity, the price can be below 100%	Natixis daily indicative price; hold to call or maturity

Pricing sensitivity (fair coupon): base 14.1% · vol ±3pts 12.9%–15.6% · corr ±0.15 13.2%–15.1% · equity drift –1/–2% 15.6%–17.4%. The 12.75% offer stays fundable in every case.

Coupon digitals

Natixis owes basket digitals struck at 100%: replicated with tight basket call spreads, delta-hedged with ETF units, constituent stocks and index futures as proxies; listed options on EWY/EWT

Vol & correlation

Coupon value moves with basket vol and correlation (fair 12.9%–15.6% for vol ±3pts): managed with listed index options and correlation trades

Quanto FX

EWT and EWY already trade in USD; 3119 (HKD, pegged), 2644 (JPY) and 00878 (TWD) need FX forwards and a quanto adjustment

Short put & funding

Natixis is long the client's put at 100% (live only if never called), hedged with basket put spreads and delta. Principal funded at 5.69%

Appendix A: pricing assumptions and methodology

Vols and expected returns from the team risk/return app (slides 4–5); correlations and dividends to refresh on Bloomberg

ETF	Weight	Vol	Div	3119 HK	EWY US	EWT US	2644 JP	00878 TT
3119 HK	35%	18%	1.0%	1.00	0.80	0.85	0.70	0.60
EWY US	20%	49%	1.2%	0.80	1.00	0.70	0.55	0.55
EWT US	15%	36%	2.6%	0.85	0.70	1.00	0.60	0.75
2644 JP	15%	26%	0.8%	0.70	0.55	0.60	1.00	0.45
00878 TT	15%	37%	7.6%	0.60	0.55	0.75	0.45	1.00

Model. Correlated GBM for each ETF, 200,000 paths, quarterly steps. Risk-neutral drift = USD funding (linear interpolation of case grid) – dividend yield. Quanto adjustment ignored (assumed hedged by the desk; to be added with FX-equity correlations).

Discounting. USD funding grid from the case rules; 5Y discount factor 0.758.

Coupon solve. Bisection on coupon so that PV = 98% (2% hedging cost and margin).

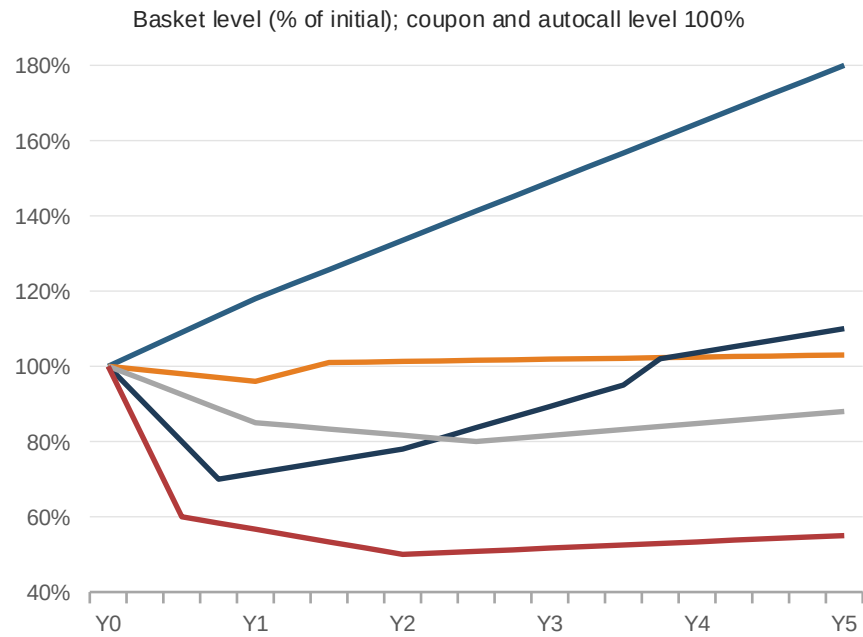
Historical back-test (to run). Monthly Bloomberg history since 2021 for all five ETFs (00878 and 3119 launched in 2020–21). Launch a hypothetical note every month, apply the same coupon, memory and autocall rules, and record coupons, life and IRR by launch year.

Variant of our note	Fair cpn
Coupon barrier 100% (base)	14.1%
Coupon barrier 95%	12.9%
Coupon barrier 90%	11.9%
First autocall at Y2	13.3%
Equity drift –1%	15.6%
Equity drift –2%	17.4%
Vol +3pts / –3pts	15.6% / 12.9%

All variants priced at 98% with the same basket and inputs. Code: `pricing/scenario2_autocall_mc.py`

Appendix B: scenario analysis, five paths for Asian markets

Hypothetical quarterly basket paths and what the USD 100mn note returns in each



Scenario	Outcome	Coupons	USD mn	IRR	Basket
AI capex boom	Called Q4 (1y)	12.8%	112.8	13.4%	–
Range-bound Asia	Called Q6 (1.5y)	19.1%	119.1	12.4%	–
2022-style tech drawdown, slow recovery	Called Q15 (3.75y)	47.8%	147.8	11.0%	–
Lost decade (sideways below par)	88% back at Y5	0.0%	88.0	-2.5%	-12%
Geopolitical shock, no recovery	55% back at Y5	0.0%	55.0	-11.3%	-45%

Reading: the note earns ~11.0%–13.4% a year whenever the basket gets back to 100%, even after a 30% drawdown: memory pays all 15 missed coupons at Q15. If the basket never recovers (lost decade, lasting shock) the note repays the basket's level, 88% or 55%.

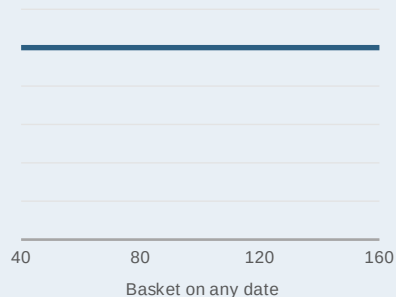
Paths are illustrative and not forecasts. 'Basket' = direct basket return at Y5 when the note is not called. Historical back-test to run on Bloomberg history (appendix).

Appendix C: how Natixis replicates the note

Replication used by the Natixis desk to price and hedge (values in % of notional, at trade date)

Principal

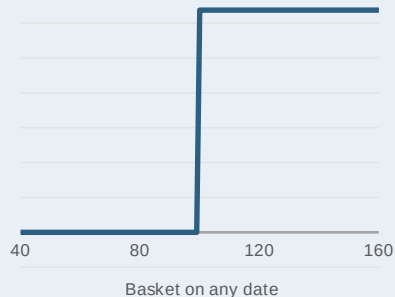
100% repaid at autocall or maturity.
Worth 89.3%



+

Coupon digitals

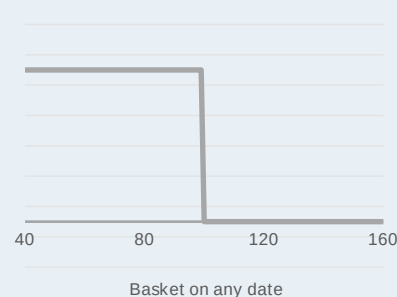
3.19% plus missed coupons on each date $\geq 100\%$. Worth 14.1% (net of autocall)



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Autocall (Natixis)

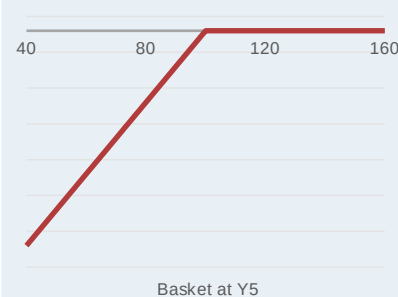
From Q4, basket $\geq 100\%$: remaining coupons cancelled, note ends. Keeps the coupon affordable



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Short put (client)

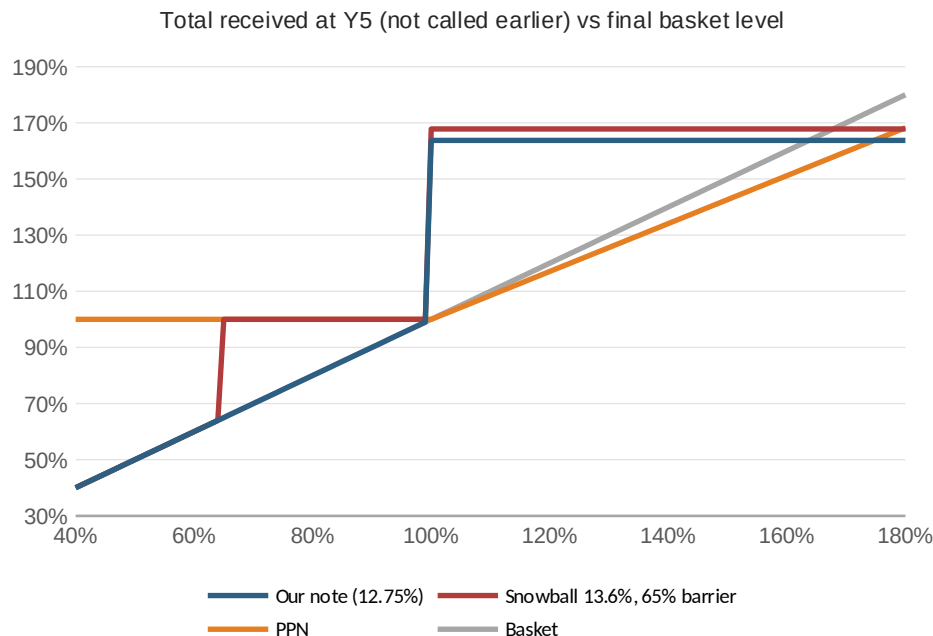
If never called, the client bears the basket's fall below 100% at Y5. Worth -6.8%



Fair value 96.5% = principal 89.3% + coupons 14.1% – short put 6.8%. Issued at 100%, leaving 3.5% for hedging and margin. The short put funds the double-digit coupon.

Appendix D: alternatives for the investment committee

Same basket: full protection, a higher-risk snowball, and a pure participation note



Option	Return	Loss risk
Classic autocallable (ours)	12.75% p.a. memory coupon	0.8% (team outlook)
100% protected autocallable	~7.9% fair coupon	0%
Snowball, 65% barrier	13.6% fair, paid at call	14.0% (pricing)
Principal-protected note	85% of basket gain at Y5	0%

Trade-off: The snowball's fair coupon is 13.6%, but it loses capital in ~14% of paths (pricing measure). Full protection pays only ~8%. The PPN pays nothing unless the basket ends above 100% after 5 years. Our note pays 12.75%, with money usually back within ~1 year on the team outlook, but takes the basket's fall if never called.

Monte Carlo, 200k paths. Vols, dividends and correlations are assumptions to refresh with Bloomberg data as of the 17 Sep 2026 trade date; discounting uses the USD funding grid.